EXCELLENCE

CLASS 11 · 2026–27

Physics | Lecture - 03

Basic Maths and Calculus

Leftover radian, then feel of trigonometry

By — Excellence Programme

3 September 2026

Topics to be covered

01 Inequalities, identities, brother angle, then the 0°–180° table for all six ratios

02 Leftover from last class, from slide 29 — radian, conversions, leftover PYQs

03 Feel of trigonometry — θ outside a ratio must be in radian

04 Compound angles A ± B, then double angle and half angle

05 Allied angles, even / odd, graphs of all six ratios 0° to 360°

06 Small-angle approximation — $\sin\theta \approx \theta$, $\tan\theta \approx \theta$, $\cos\theta \approx 1$

Inequalities from Lectures 01 and 02

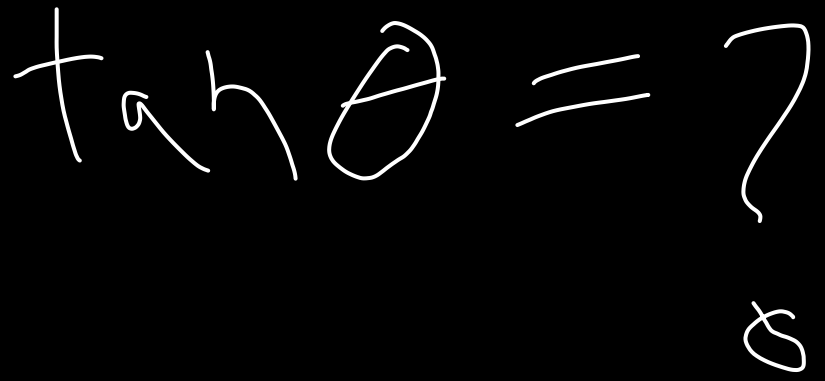
H is the longest side, so P ≤ H and B ≤ H. That is the whole range.

$$-1 \leq \sin\theta \leq 1$$
$$-1 \leq \cos\theta \leq 1$$

tan θ has no finite range
tan 90° is not defined

sin θ = P/H H > P so |sin θ| ≤ 1

If a question writes sin θ = 2, stop. No real angle. Do not open the identity.



Reciprocals, and 0° to 90°

$|\text{cosec}\ \theta| \geq 1$ **$|\sec\theta| \geq 1$**

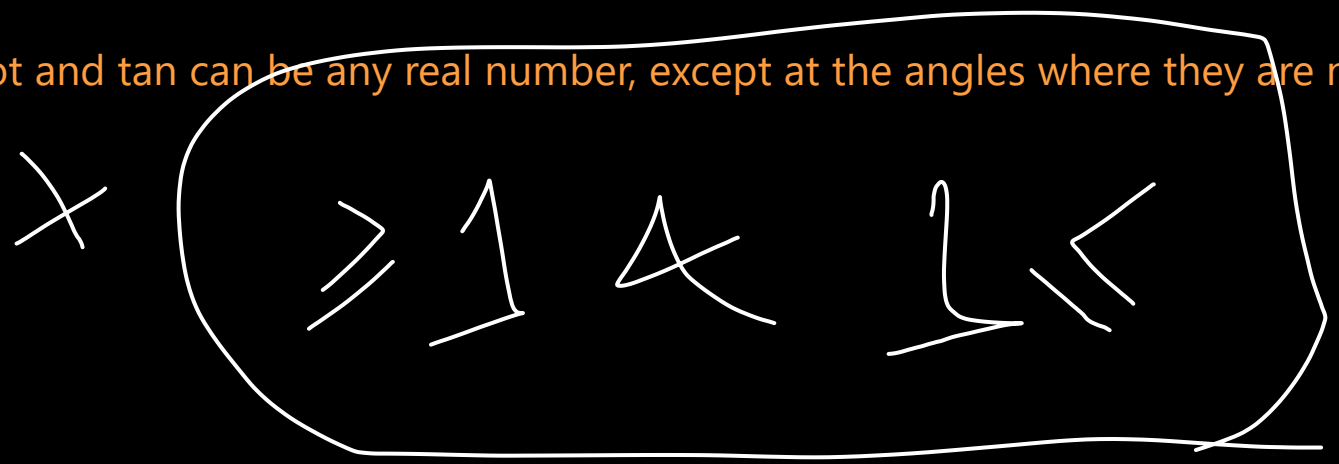
When they are defined. Cosec is 1/sin, sec is 1/cos. If |sin| ≤ 1, then |cosec| cannot be 1/2.

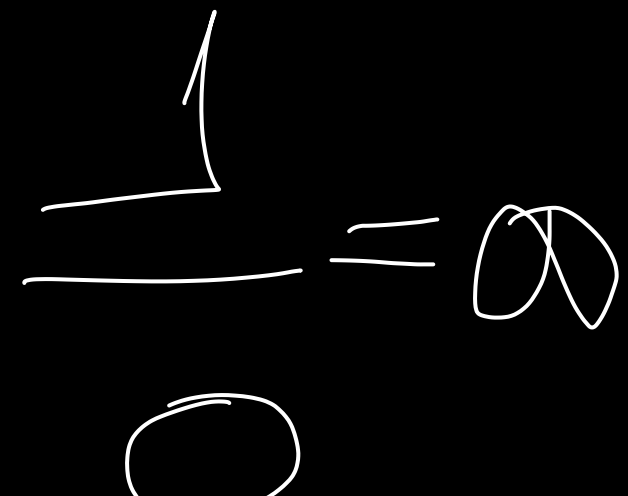
cosec 0° and cosec 180° are not defined **cot 0° and cot 180° are not defined**

sec 90° is not defined **tan 90° is not defined**

0° → 90°: **sin increases** **cos decreases** **tan increases**

cot and tan can be any real number, except at the angles where they are not defined.





The three identities — one Pythagoras

Start from $P^2 + B^2 = H^2$. Divide by H^2, by B^2, by P^2.

$$\sin^2\theta + \cos^2\theta = 1$$
$$1 + \tan^2\theta = \sec^2\theta$$
$$1 + \cot^2\theta = \operatorname{cosec}^2\theta$$

If you keep only one line: $\sin^2\theta + \cos^2\theta = 1$

Brother angle in the second quadrant

An angle past 90° uses a brother angle in the first quadrant. ASTC gives the sign.

sin (180° − θ) = sin θ
cos (180° − θ) = − cos θ
tan (180° − θ) = − tan θ

Example: sin 120° = sin 60° = √3/2. cos 120° = − cos 60° = −1/2.

120° = 180° − 60°

$\sin(150) = 180 - 30 = \sin 30$

$= \frac{1}{2}$

All six ratios from 0° to 90°

θ	0°	30°	45°	60°	90°
sin	0	1/2	1/√2	√3/2	1
cos	1	√3/2	1/√2	1/2	0
tan	0	1/√3	1	√3	undef.
cot	undef.	√3	1	1/√3	0
sec	1	2/√3	√2	2	undef.
cosec	undef.	2	√2	2/√3	1

tan 90° and sec 90° are not defined. cosec 0° and cot 0° are not defined.

All six ratios from 90° to 180°

θ	90°	120°	135°	150°	180°
sin	1	√3/2	1/√2	1/2	0
cos	0	−1/2	−1/√2	−√3/2	−1
tan	undef.	−√3	−1	−1/√3	0
cot	0	−1/√3	−1	−√3	undef.
sec	undef.	−2	−√2	−2/√3	−1
cosec	1	2/√3	√2	2	undef.

Second quadrant: only sin and cosec are positive. cot 180° and cosec 180° are not defined.

Last class — ASTC

A S T C → All sin tan cos

First quadrant: all positive. Reciprocals take the same sign as the ratio.

$\sin(180° - \theta) = \sin\theta$ **$\cos(180° - \theta) = -\cos\theta$**

$\tan 90°$ is not defined. **$-1 \le \sin\theta \le 1$** **$-1 \le \cos\theta \le 1$**

Last class — table and 37° / 53°

sin 37° = 3/5 cos 37° = 4/5 tan 37° = 3/4

sin 53° = 4/5 cos 53° = 3/5 tan 53° = 4/3

0° → 90°: sin increases, cos decreases, tan increases.

$\sin^2 40° + \cos^2 40° = 1$

Last class stopped at the break. Radian was not taught. That leftover is first today.

Last class — height and distance

Elevation: $h = x \tan \theta$

Depression: $D = (h_1 - h_2) / \tan \theta$

Sketch first. Opposite is the height, or the difference of heights, never the sum.

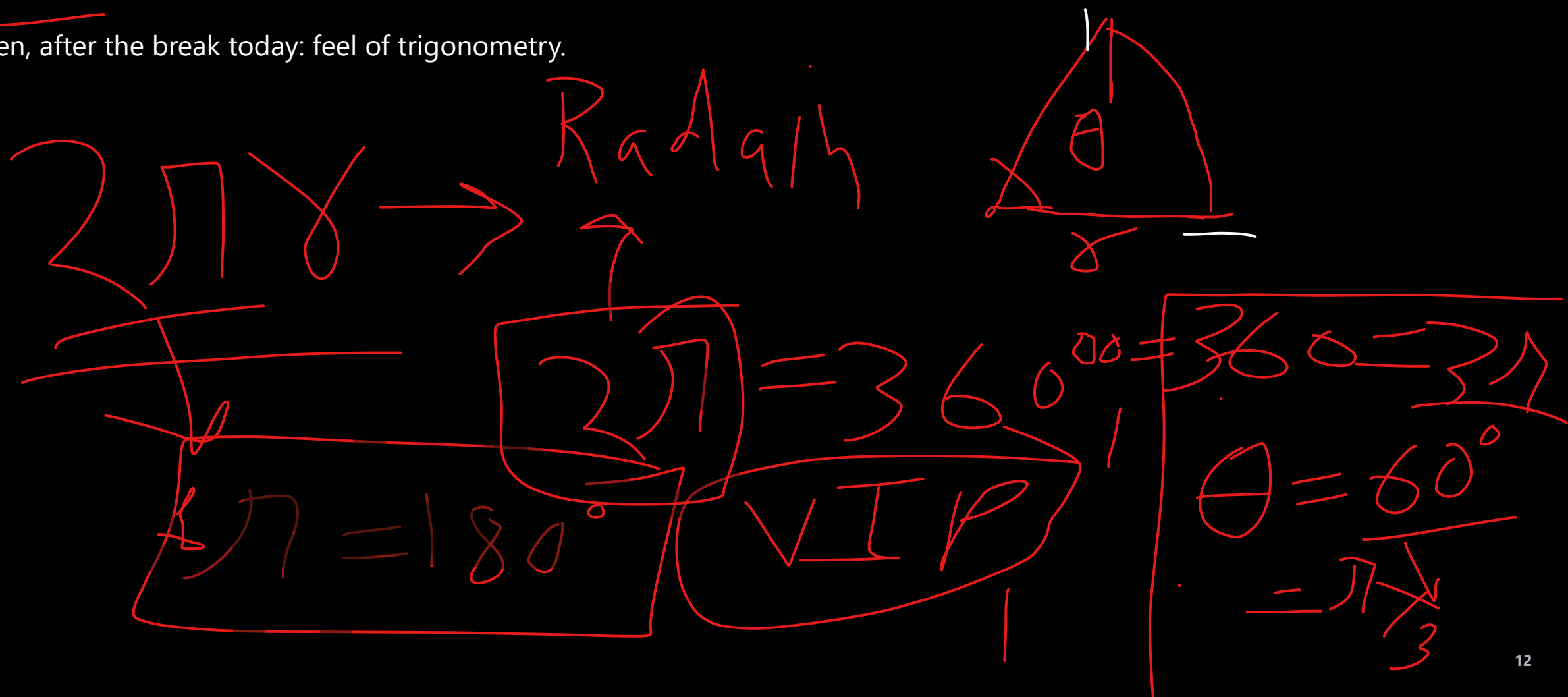
We stopped at slide 29. First today: leftover radian, conversions, leftover questions.

Last class stopped here — radian starts now

Last class ended at the break. The next board was radian. We do that leftover first.

Arc length → 180° = π rad → conversions → leftover questions

Then, after the break today: feel of trigonometry.



Arc length

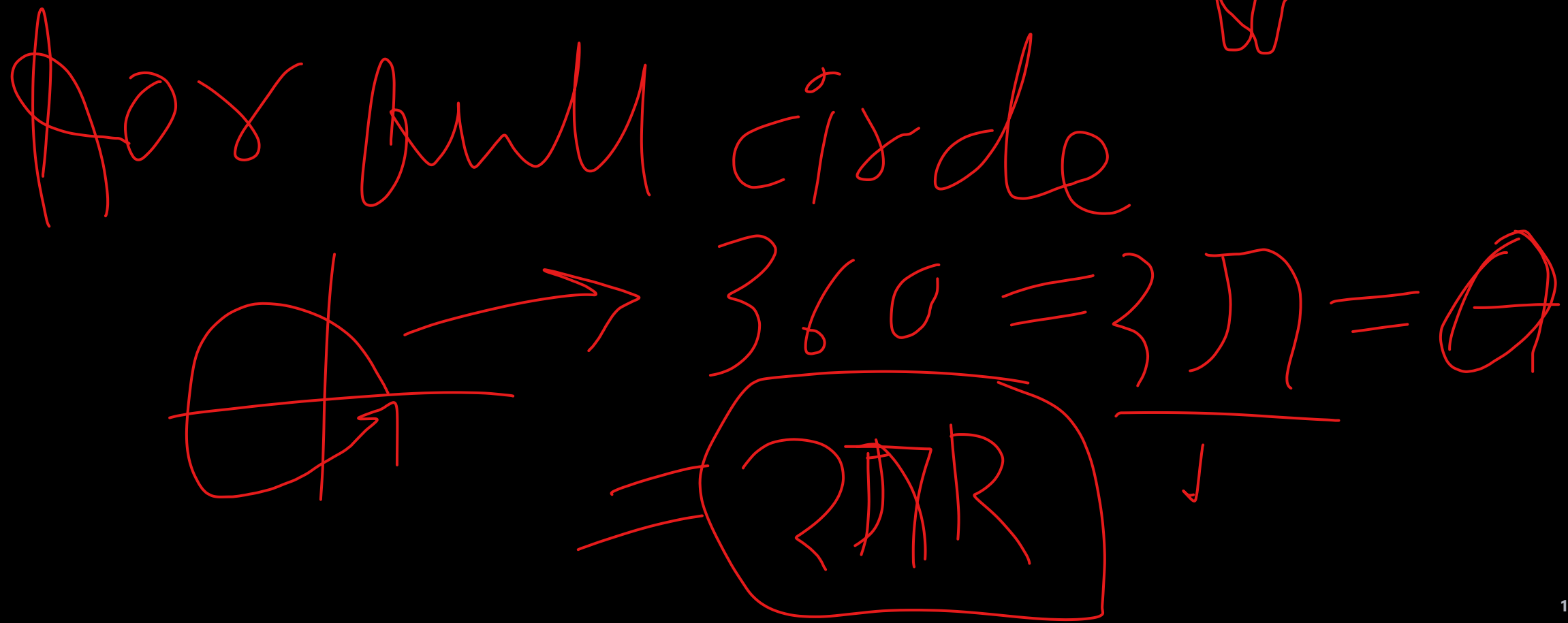
Arc length = R θ

θ must be in radian. Never put 90° into this formula.

Practical unit of angle: degree. SI unit of angle: radian.

+θ anti-clockwise −θ clockwise

Arc = R θ



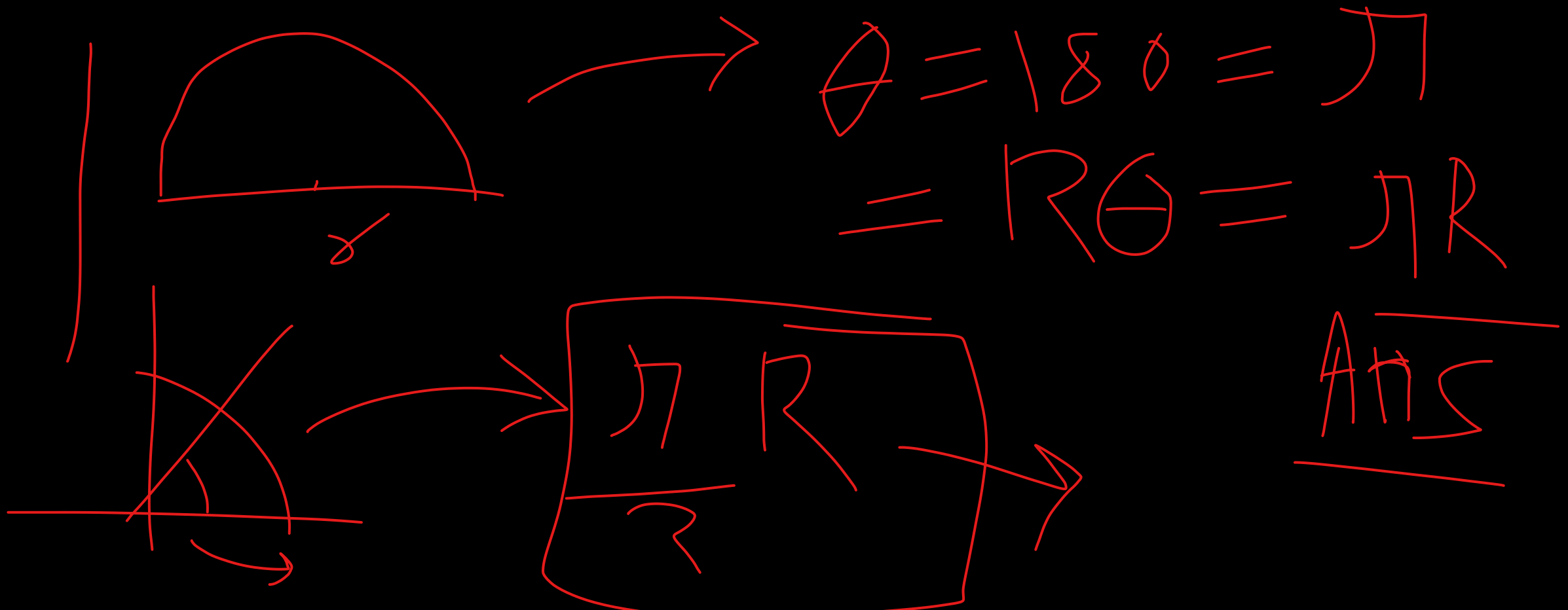
The one line: 180° = π rad

Lock this. Every conversion starts here.

180° = π rad

Never write π = 180°. Write π rad = 180°.

Wherever you see π rad, put 180°.



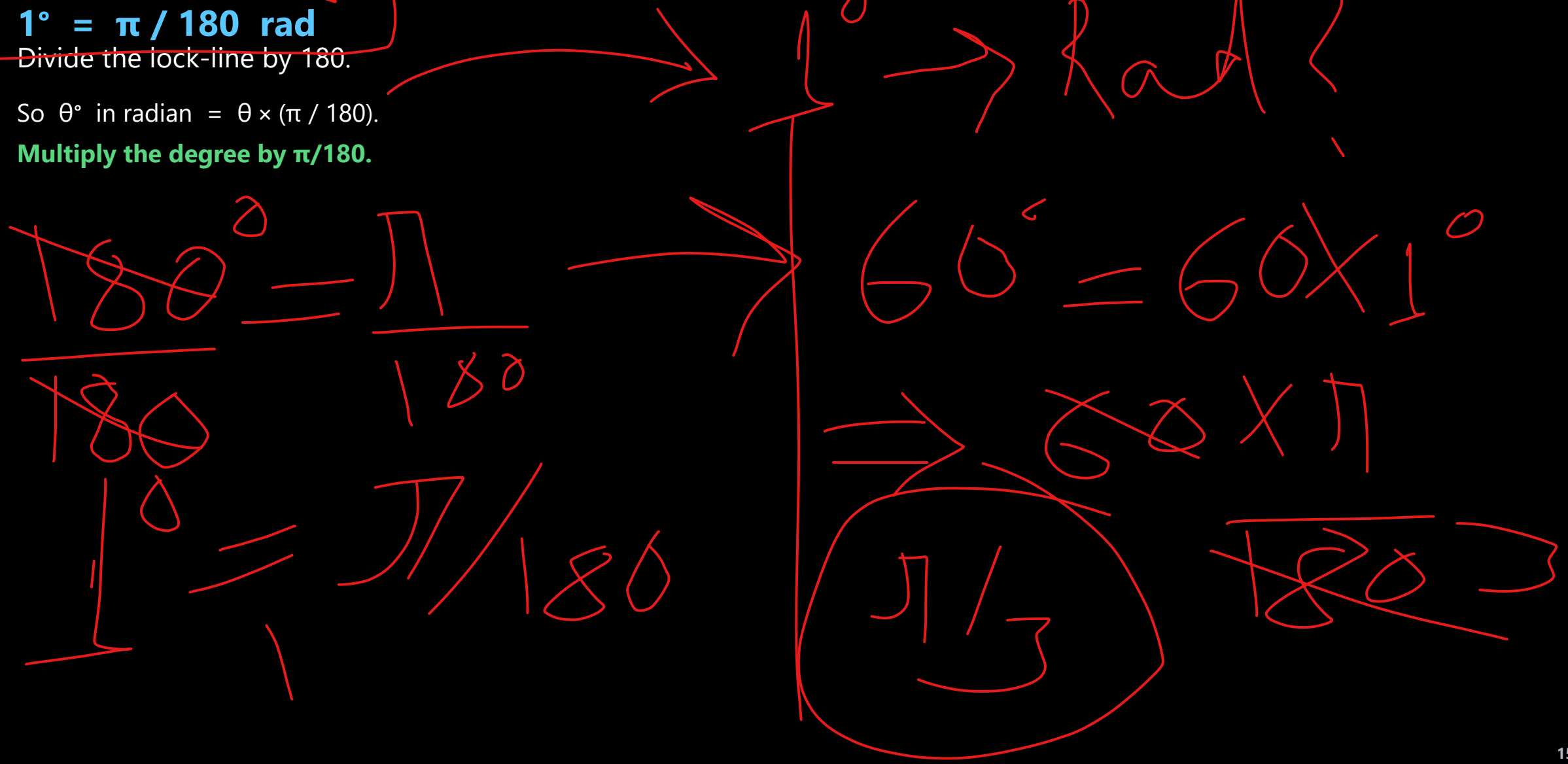
Degree → radian

1° = π / 180 rad

Divide the lock-line by 180.

So θ° in radian = θ × (π / 180).

Multiply the degree by π/180.



Radian → degree

1 rad = 180° / π

Divide the lock-line by π.

So θ rad in degree = θ × (180 / π).

Multiply the radian by 180/π.

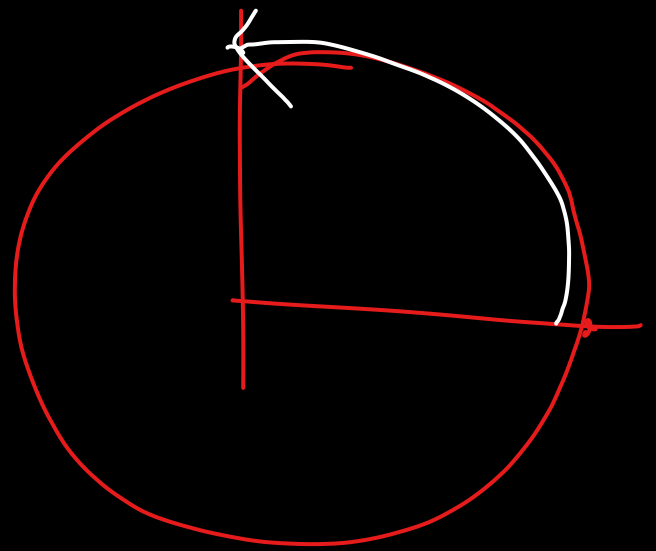


Arc on a circle

An object moves on the circumference of a circle of radius R. If the object turns through 90°, find the total distance moved by the object.

$\theta = 90 = \frac{\pi}{2}$

$\frac{\pi}{2} \times R$



Object turns 90° on a circle of radius R

An object moves on the circumference of a circle of radius R and turns through 90°. Find the distance.

Wrong: Distance = R × 90°

Right: Distance = R × (π/2 rad) = πR/2

distance = path = arc = Rθ

Convert 90° first: 90° = π/2 rad

πR / 2

First conversion

Convert π/3 radian into degree.

$$\frac{\pi}{3} \times \text{RoA} = \frac{\pi}{3} \times \frac{180}{\pi}$$

$$= 60^\circ$$

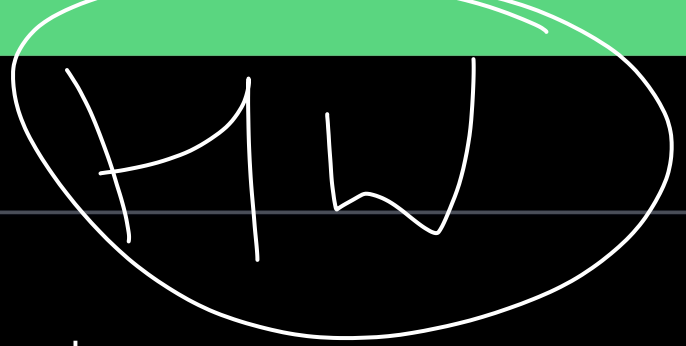
π / 3 rad

Wherever you see π rad, put 180°.

π / 3 rad = 180° / 3 = 60°

π / 3 rad = 60°

Convert these now



Convert the following into degree or radian, as required.

(i) π/2 rad

(ii) π/4 rad

(iii) π/6 rad

(iv) 45°

(v) 90°

(vi) 60°

The six conversions from the board

$\pi/2$ rad = 90°

$\pi/6$ rad = 30°

90° = $\pi/2$ rad

$\pi/4$ rad = 45°

45° = $\pi/4$ rad

60° = $\pi/3$ rad

More radian into degree

Convert the following radian measures into degree.

(i) $5\pi/6$
(ii) $2\pi/3$
(iii) $4\pi/3$
(iv) $3\pi/5$
(v) 5π
(vi) $2\pi/5$
(vii) $7\pi/12$
(viii) $5\pi/2$
(ix) 3π

Radian into degree

$5\pi/6 = 150°$

$2\pi/3 = 120°$

$4\pi/3 = 240°$

$3\pi/5 = 108°$

$5\pi = 900°$

$2\pi/5 = 72°$

$7\pi/12 = 105°$

$5\pi/2 = 450°$

$3\pi = 540°$

Degree into radian

Convert the following into radian.

(i) $120°$
(ii) $30°$
(iii) $150°$
(iv) $270°$
(v) $135°$
(vi) π

(vi) is written π, not $\pi°$. Do not treat it as a degree.

Degree into radian

120° = 2π/3 rad

270° = 3π/2 rad

30° = π/6 rad

135° = 3π/4 rad

150° = 5π/6 rad

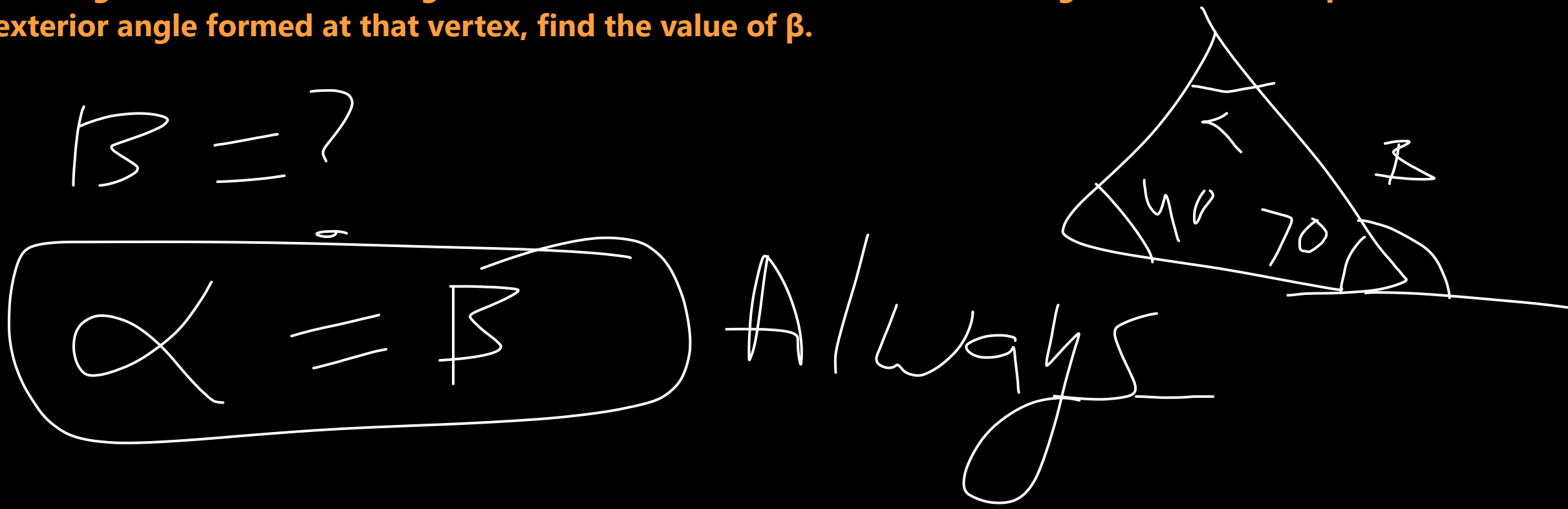
π is already radian

Wrong: π° treated as π rad

Right: π is already radian. π° would be π × (π/180) = $\pi^2/180$ rad

Find β

A triangle has two interior angles 40° and 70°. The side next to the 70° angle is extended. If β is the exterior angle formed at that vertex, find the value of β.



Exterior angle β

$40° + 70° + \alpha = 180°$ $\alpha = 70°$

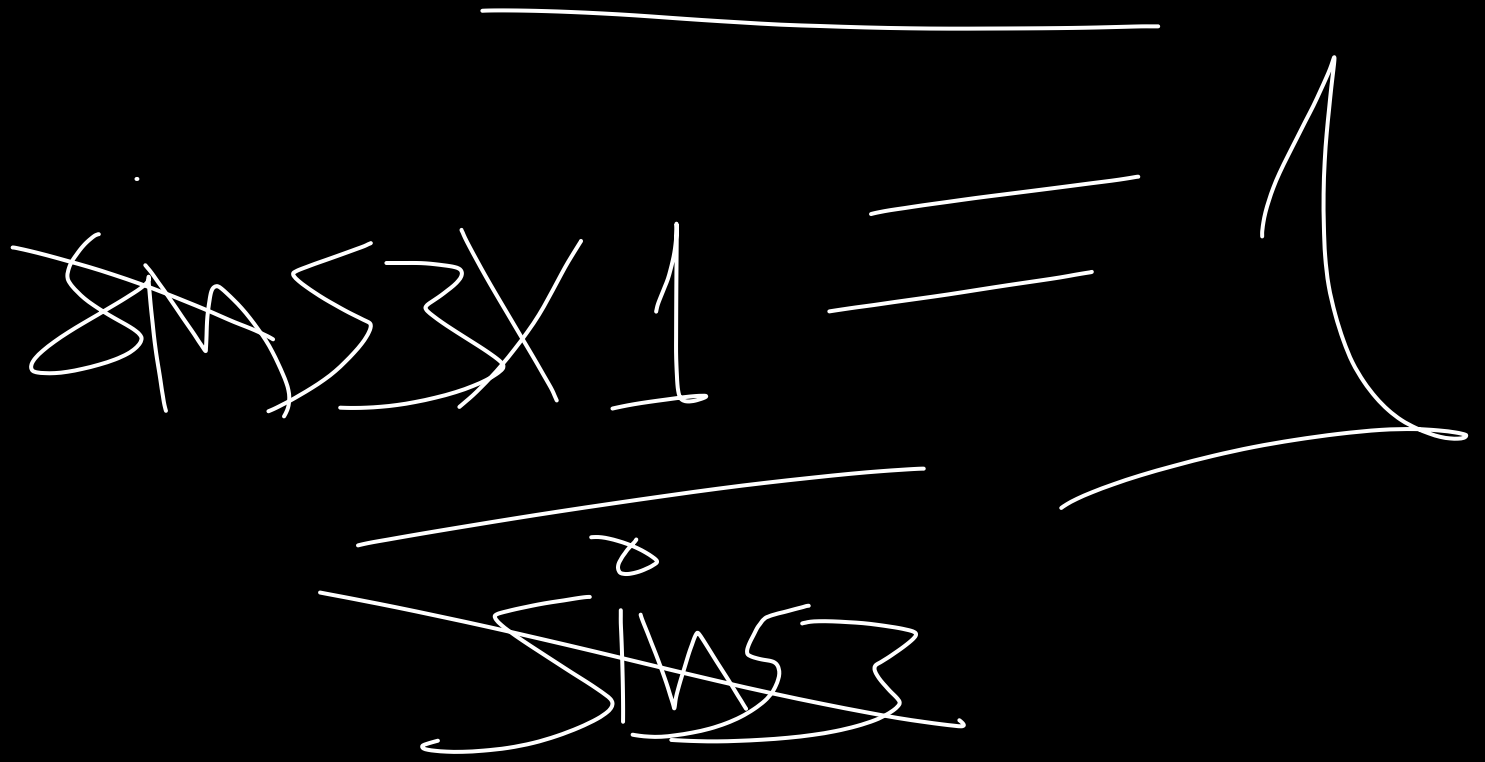
$\beta = 180° - \alpha = 110°$

$\beta = 110°$

Shortcut: an exterior angle equals the sum of the two remote interior angles. 40° + 70° = 110°.

Product of a ratio and its reciprocal

Find the value of sin 53° × cosec 53°.



$\sin 53° \times \operatorname{cosec} 53°$

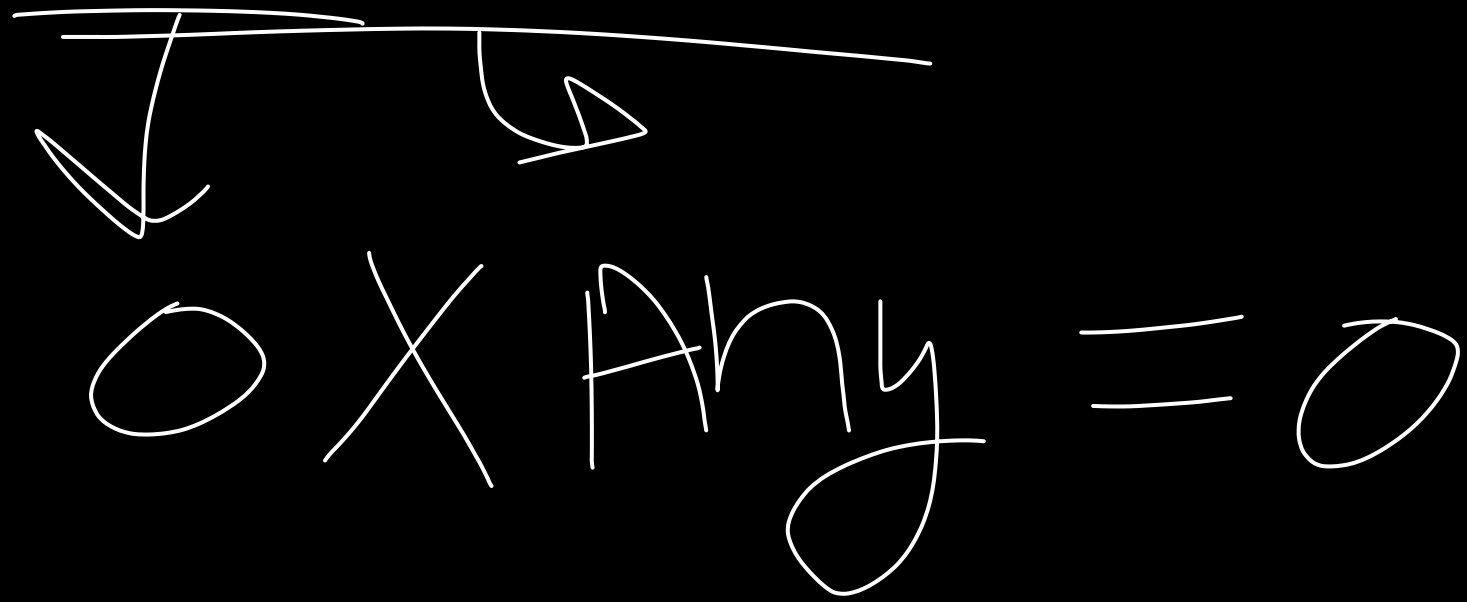
$\sin \theta \times \operatorname{cosec} \theta = 1$

The angles are the same, so the identity is enough. Do not look up 53°.

$\sin 53° \times \operatorname{cosec} 53° = 1$

Product of three values

Find the value of $\sin 180° \cdot \cos 135° \cdot \tan 120°$.



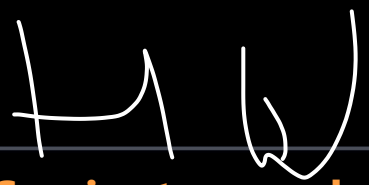
$\sin 180° \cdot \cos 135° \cdot \tan 120°$

$\sin 180° = 0$

Zero times anything is zero. You do not need $\cos 135°$ or $\tan 120°$ after that.

0

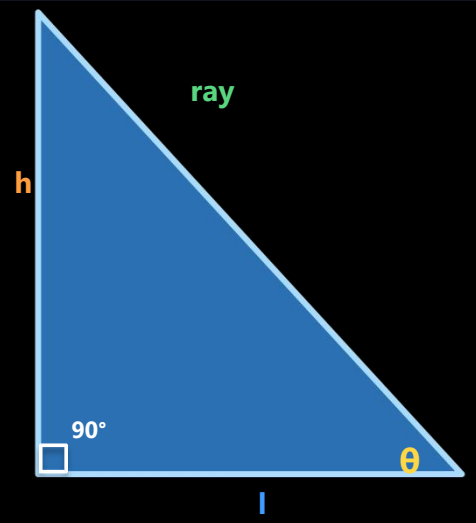
Shadow of a pole



A vertical pole of height h casts a shadow of length l when the Sun is at an angle θ above the horizon. Which of the following expresses the shadow length l in terms of h and θ?

1. $l = h \tan \theta$
2. $l = h / \tan \theta$
3. $l = h \sin \theta$
4. $l = h / \sin \theta$

Shadow of a pole

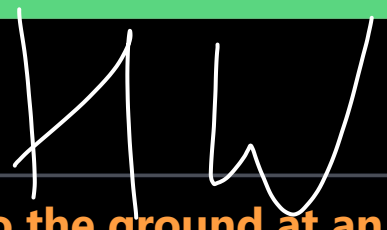


$\tan \theta = h / l$

$l = h / \tan \theta = h \cot \theta$

$l = h / \tan \theta$

Wire from the top of a pole



A pole of height h has a wire attached from the top to the ground at an angle θ. If the wire breaks and falls to the ground without changing length, find the horizontal distance x from the base of the pole where the far end of the wire then lies.

1. $x = h \cot \theta$
2. $x = h \tan \theta$
3. $x = h \sin \theta$
4. $x = h \sec \theta$

Wire falls without changing length

Wire length $l = h / \sin\theta = h \operatorname{cosec}\theta$

Original base $b = h / \tan\theta = h \cot\theta$

After the fall, the wire lies on the ground. $x = l - b$

$$x = h(\operatorname{cosec}\theta - \cot\theta)$$

h cot θ is the original base, not the far end after the wire falls. None of the four options is x.

Shadow on sloping ground

A vertical pole of height h = 10 m stands on ground that slopes upwards at a constant angle α = 30° with the horizontal. If the Sun's angle of elevation above the horizontal is θ = 60°, what is the length of the shadow cast by the pole on the sloping ground?

1. 5 m
2. 10 m
3. $10\sqrt{3}$ m
4. $10 / (\sqrt{3} - 1)$ m

Shadow on sloping ground

Let the shadow on the slope be l. Rise of the slope over that length is a. Horizontal run is b.

$a = l \sin 30° = l/2$ $\quad b = l \cos 30° = l\sqrt{3}/2$

$\tan 60° = (10 + a)/b \quad \sqrt{3} = (10 + l/2)/(l\sqrt{3}/2)$

$3l/2 = 10 + l/2 \quad l = 10$

$l = 10$ m

BREAK

Leftover radian is done. Next: feel of trigonometry

Length prefixes you will meet in physics

Write this list once. You will use it from Units and Measurements onward.

1 m = 100 cm
1 fm = 10^{-15} m
1 pm = 10^{-12} m
1 nm = 10^{-9} m
1 μm = 10^{-6} m

1 mm = 10^{-3} m
1 cm = 10^{-2} m
1 km = 10^{3} m
1 Mm = 10^{6} m
1 Å = 10^{-10} m

Å is ångström. Nuclear sizes are in fm. Light wavelengths are in nm or Å.

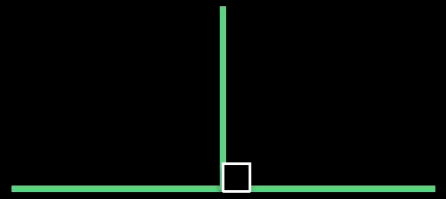
Lines — parallel, orthogonal, antiparallel

These three words come back in vectors. Lock the picture today.



Parallel

θ = 0°



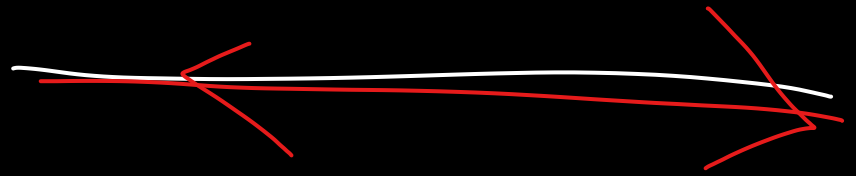
Orthogonal (perpendicular)

θ = 90°



Antiparallel

θ = 180°



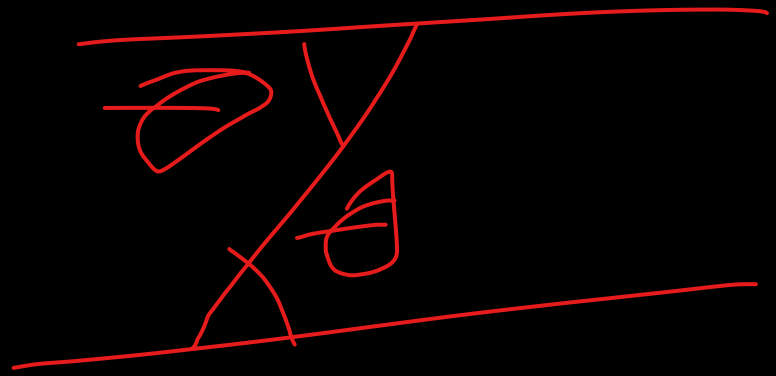
Complementary, acute, obtuse, alternate

Complementary: $\alpha + \beta = 90°$ so $\alpha = 90° - \beta$

In a right triangle the two acute angles are complementary. That is why $\sin(90° - \theta) = \cos\theta$.

Acute: $0° \leq \theta < 90°$ **Obtuse:** $90° < \alpha \leq 180°$

Alternate angles: a transversal cuts two parallel lines. The inner opposite angles are equal.



θ outside a ratio is not a degree

If y = sin θ / θ, find the value of y when θ = 30°.

Write the line. Do not divide by 30. The next board is the trap.

The denominator must be in radian

Wrong: y = sin 30° / 30°

Right: y = sin 30° / (π/6)

30° = π/6 rad **sin 30° = 1/2**

y = (1/2) ÷ (π/6) = (1/2) × (6/π) = 3/π

If θ is not inside sin, cos or tan, convert it to radian first.

(a/b) ÷ (c/d) = ad / bc.

Compound angles — sine and cosine

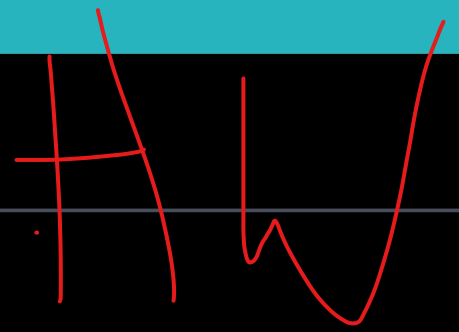
A and B are angles. Copy these four lines. Practice them ten times on a rough page.

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$
$$\sin(A - B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$
$$\cos(A - B) = \cos A \cos B + \sin A \sin B$$

Cosine of a sum has a minus. Sine of a sum has a plus. Do not swap those two signs.

Compound angles — tangent

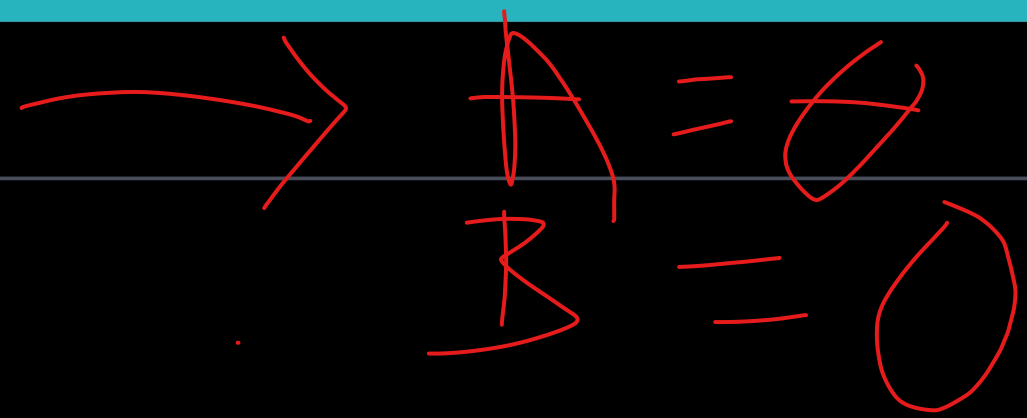


$$\tan(A + B) = (\tan A + \tan B) / (1 - \tan A \tan B)$$
$$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$$

The minus in the denominator belongs to the sum. The plus belongs to the difference.

If $\tan A \tan B = 1$, then $\tan(A + B)$ is not defined.

Double angle from sin (A + B)



Put A = B = θ in the sine-of-a-sum line.

sin (θ + θ) = sin θ cos θ + cos θ sin θ

sin 2θ = 2 sin θ cos θ

Now let 2θ = α, so θ = α/2.

sin α = 2 sin(α/2) cos(α/2)

Sin(x)*Cos(x) + Sin(x)*cos(x)

2*Sin(x)*Cos(x)

Double angle from cos (A + B)

Put A = B = θ in the cosine-of-a-sum line.

$\cos 2\theta = \cos^2\theta - \sin^2\theta$

$2\sin^2\theta = 1 - \cos 2\theta$

$\sin\theta = \sqrt{[(1 - \cos 2\theta) / 2]}$

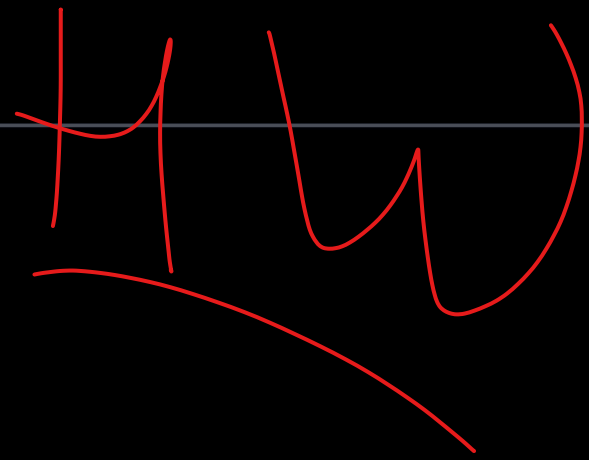
sin^2 + cos^2 = 1
sin^2 = 1- cos^2
cos^2 = 1- sin^2

$\cos 2\theta = 1 - 2\sin^2\theta$

Half angle: let $2\theta = \alpha$. Then $\cos\alpha = \cos^2(\alpha/2) - \sin^2(\alpha/2)$.

cos^2 - sin^2 = cos^2 -1 +cos^2
cos^2 - sin^2 = 2*cos^2 -1
cos^2 - sin^2 = 2*(1- sin^2) -1
cos^2 - sin^2 = 2- 2sin^2 -1
cos^2 - sin^2 = 1- 2sin^2

Find these four values



Find the value of each of the following.

(1) $\sin 120°$

(2) $\cos 150°$

(3) $\sin 240°$

(4) $\cos 120°$

sin 120° — two routes, one value

Write 120° as a sum you already know.

sin (90° + 30°) = sin 90° cos 30° + cos 90° sin 30°

= 1 · (√3/2) + 0 = √3/2

Same value from the double-angle line: sin(2 × 60°) = 2 sin 60° cos 60° = √3/2.

sin 120° = √3/2

cos 150° and cos 120°

$\cos 150° = \cos(90° + 60°) = \cos 90° \cos 60° - \sin 90° \sin 60°$

$= 0 - 1 \cdot (\sqrt{3}/2) = -\sqrt{3}/2$

$\cos 150° = -\sqrt{3}/2$

$\cos 120° = \cos(90° + 30°) = -\sin 30° = -1/2$

$\cos 120° = -1/2$

sin 240°

$$\sin 240° = \sin(180° + 60°) = \sin 180° \cos 60° + \cos 180° \sin 60°$$

$$= 0 + (-1) \cdot (\sqrt{3}/2) = -\sqrt{3}/2$$

$\sin 240° = -\sqrt{3}/2$

240° is in the third quadrant. Sine is negative there. The compound line already gives the sign.

Allied angles — sine

Simplify each of the following.

(5) $\sin(90° - \theta)$

(6) $\sin(90° + \theta)$

(7) $\sin(180° - \theta)$

(8) $\sin(270° + \theta)$

$\sin(90° - \theta)$ and $\sin(90° + \theta)$

$\sin(90° - \theta) = \sin 90° \cos\theta - \cos 90° \sin\theta = 1 \cdot \cos\theta - 0 = \cos\theta$

$\sin(90° - \theta) = \cos\theta$

$\sin(90° + \theta) = \sin 90° \cos\theta + \cos 90° \sin\theta = \cos\theta$

$\sin(90° + \theta) = \cos\theta$

Both become cosine. The plus or minus next to 90° does not change sine into minus cosine here.

$\sin(180° - \theta)$ and $\sin(270° + \theta)$

$\sin(180° - \theta) = 0 \cdot \cos\theta - \sin\theta \cdot (-1) = \sin\theta$

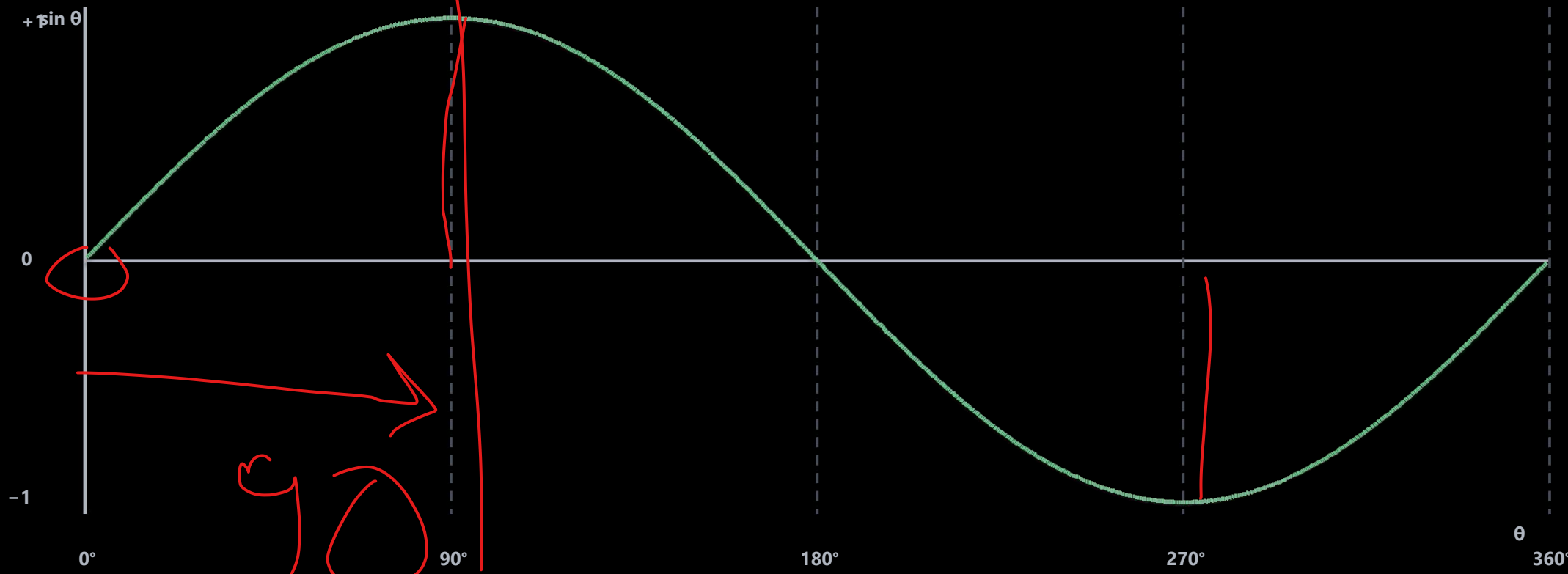
$\sin(180° - \theta) = \sin\theta$

$\sin(270° + \theta) = \sin 270° \cos\theta + \cos 270° \sin\theta = (-1)\cos\theta + 0$

$\sin(270° + \theta) = -\cos\theta$

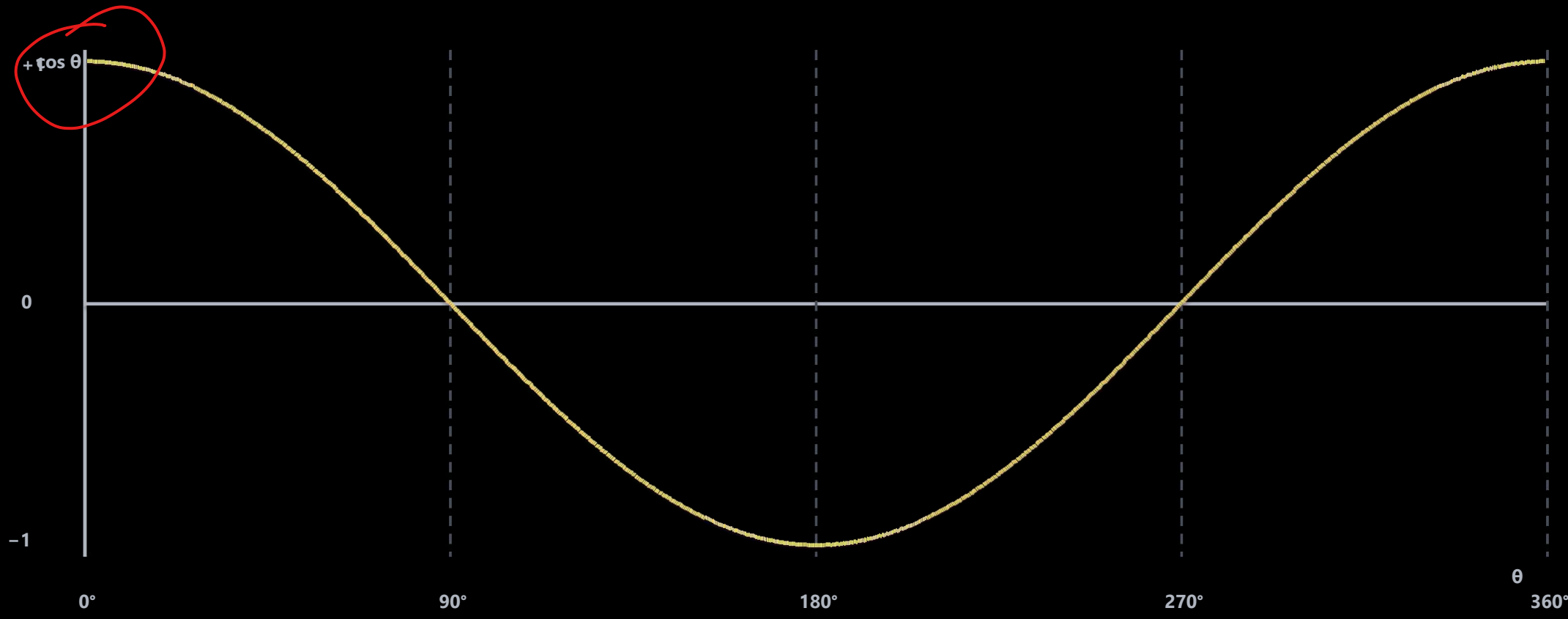
sin θ from 0° to 360°

sin 0° = sin 180° = sin 360° = 0 sin 90° = +1 sin 270° = −1



cos θ from 0° to 360°

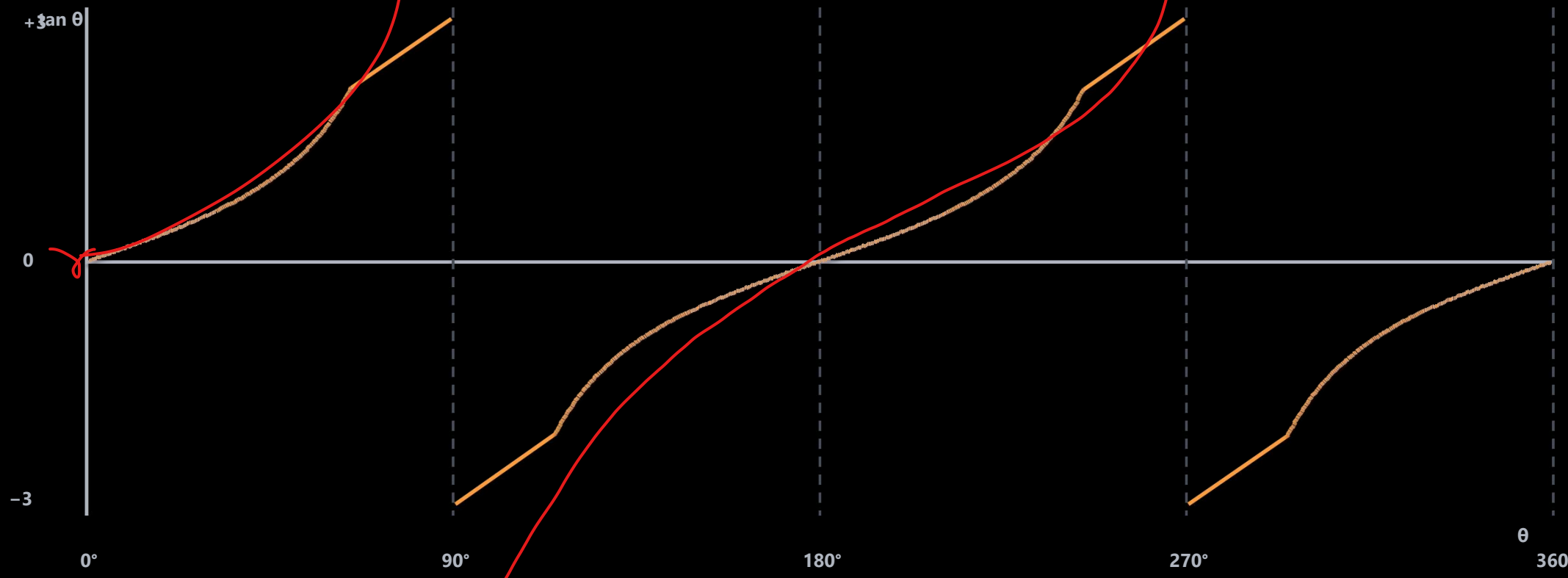
cos 0° = +1 cos 90° = 0 cos 180° = −1 cos 270° = 0



tan θ from 0° to 360°

tan 0° = tan 180° = tan 360° = 0

tan 90° and tan 270° are not defined

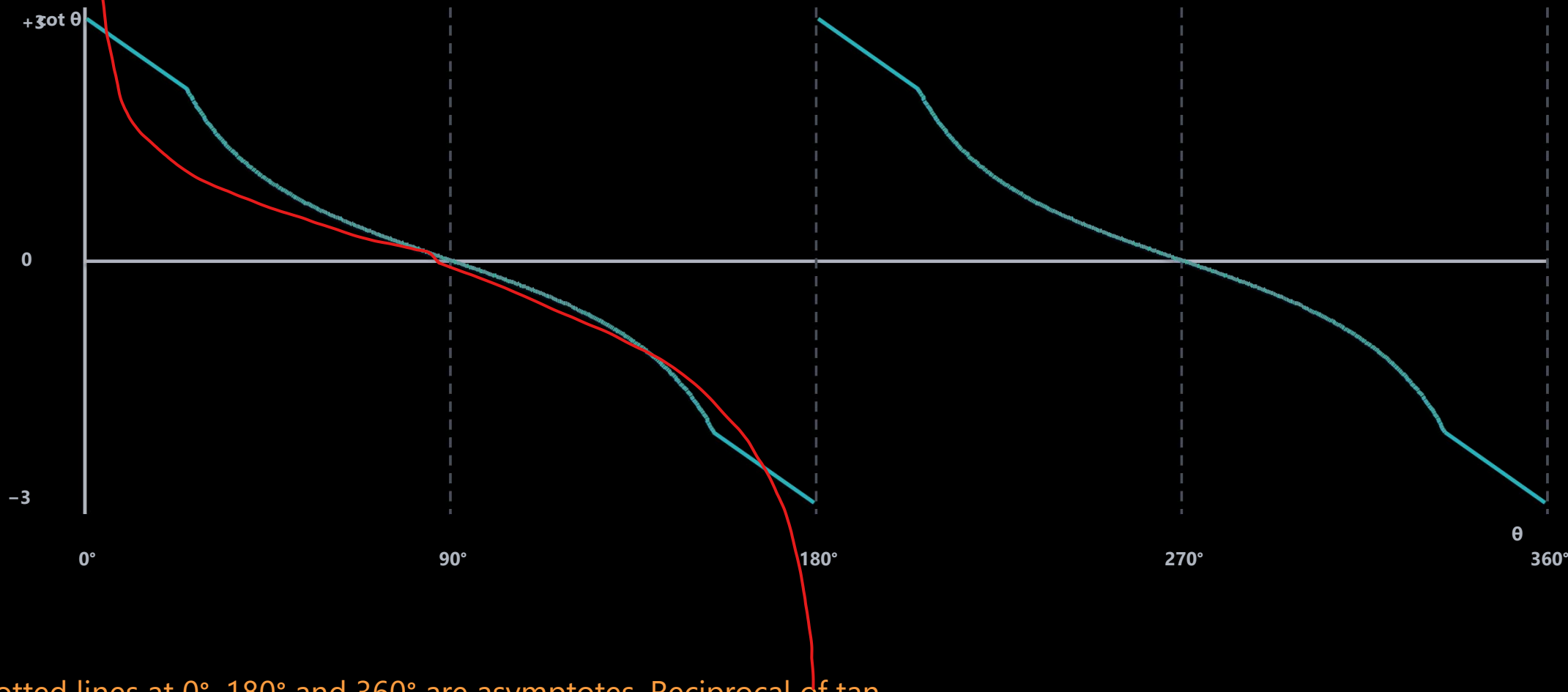


Dotted lines at 90° and 270° are asymptotes. The graph never crosses them.

cot θ from 0° to 360°

cot 90° = cot 270° = 0

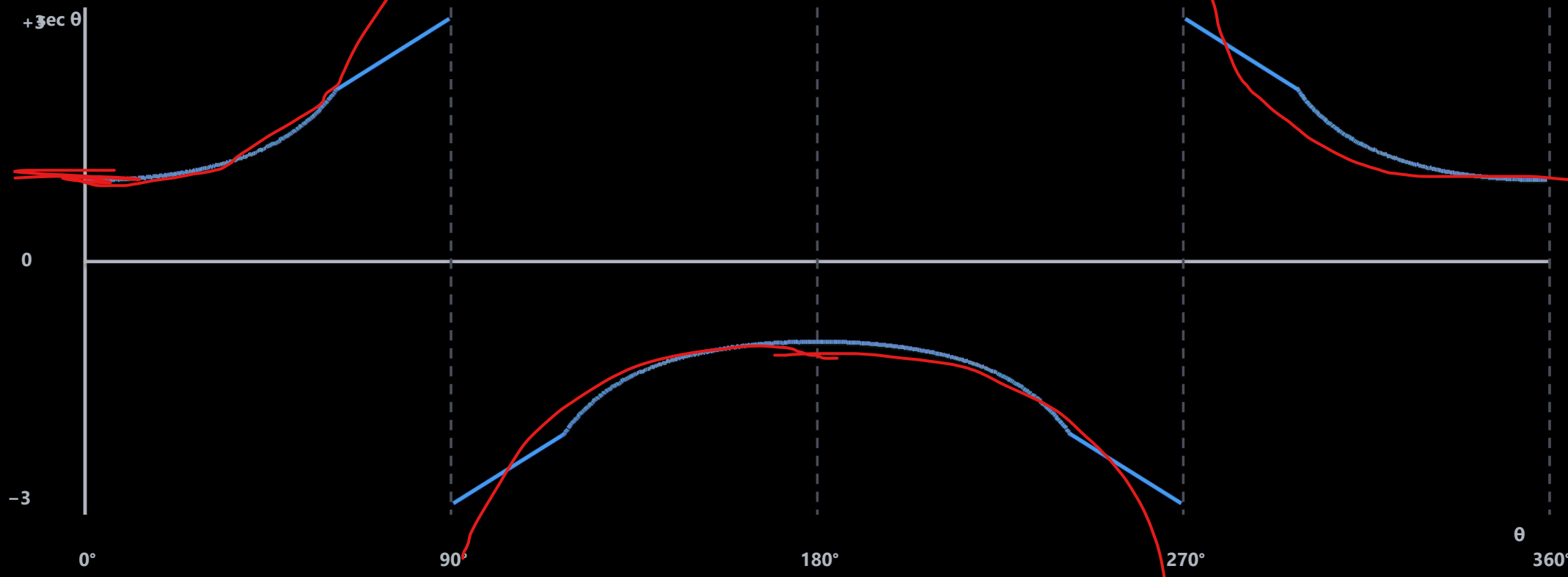
cot 0°, cot 180° and cot 360° are not defined



Dotted lines at 0°, 180° and 360° are asymptotes. Reciprocal of tan.

sec θ from 0° to 360°

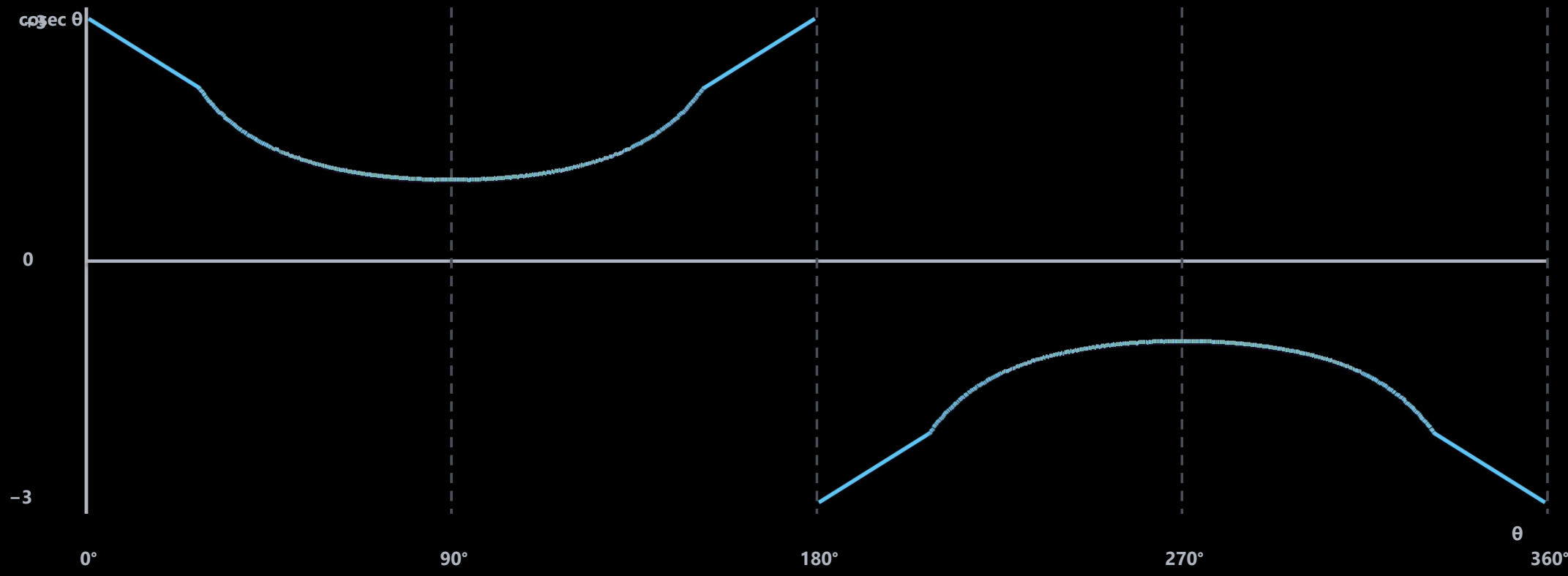
sec 0° = sec 360° = +1 sec 180° = −1 sec 90° and sec 270° are not defined



|sec θ| ≥ 1. The graph never sits between −1 and +1.

cosec θ from 0° to 360°

cosec 90° = +1 **cosec 270° = −1** **cosec 0°, 180° and 360° are not defined**



|cosec θ| ≥ 1. Reciprocal of sine. Never write cosec 0°.

Negative angles

cos 360° = 1 and sin 360° = 0. Put those into the compound lines.

cos (−θ) = cos θ

sin (−θ) = − sin θ

tan (−θ) = − tan θ

even function
odd functions

Example: cos (−30°) = cos 30° = √3/2. Also sin (360° + θ) = sin θ.

Negative angles

Find the value of each of the following.

(1) $\sin(-30°)$

(2) $\cos(-60°)$

(3) $\sin(-120°)$

(4) $\cos(-120°)$

(5) $\sin(-150°)$

Negative angles — even and odd

$\sin(-30°) = -\sin 30° = -1/2$

$\cos(-60°) = +\cos 60° = +1/2$

$\sin(-120°) = -\sin 120° = -\sqrt{3}/2$

$\cos(-120°) = +\cos 120° = -1/2$

$\sin(-150°) = -\sin 150° = -1/2$

Cosine keeps the sign of the positive angle. Sine flips. Do not flip cosine.

NEXT

37° / 53° splits → sin 15°, 75°, 105° → small angle → tonight

Split using 90° or 180°, then 37° or 53°

Write each as a compound of 90° or 180° with 37° or 53°, then find the value.

(1) sin 127°

(2) cos 143°

(3) tan 143°

Last class: sin 37° = 3/5, cos 37° = 4/5, tan 37° = 3/4.
sin 53° = 4/5, cos 53° = 3/5, tan 53° = 4/3.

127° and 143°

$\sin 127° = \sin(90° + 37°) = \cos 37° = 4/5$

$\sin 127° = 4/5$

$\cos 143° = \cos(90° + 53°) = -\sin 53° = -4/5$

$\tan 143° = \tan(180° - 37°) = -\tan 37° = -3/4$

$\cos 143° = -4/5$ $\quad$ $\tan 143° = -3/4$

Angles that are sums of 30°, 45°, 60°

Find the value of each of the following.

(1) $\sin 105° = \sin(60° + 45°)$

(2) $\sin 75° = \sin(45° + 30°)$

(3) $\sin 15° = \sin(45° - 30°)$

Use the sine-of-a-sum line. Keep √2 and √3 until the last step.

sin 105°

$\sin(60° + 45°) = \sin 60° \cos 45° + \cos 60° \sin 45°$

$= (\sqrt{3}/2)(1/\sqrt{2}) + (1/2)(1/\sqrt{2})$

$= (\sqrt{6} + \sqrt{2}) / 4$

$\sin 105° = (\sqrt{6} + \sqrt{2}) / 4$

sin 75° and sin 15°

$\sin 75° = \sin(45° + 30°) = (\sqrt{6} + \sqrt{2}) / 4$

$\sin 75° = (\sqrt{6} + \sqrt{2}) / 4$

$\sin 15° = \sin(45° - 30°) = \sin 45° \cos 30° - \cos 45° \sin 30°$

$= (1/\sqrt{2})(\sqrt{3}/2) - (1/\sqrt{2})(1/2) = (\sqrt{6} - \sqrt{2}) / 4$

$\sin 15° = (\sqrt{6} - \sqrt{2}) / 4$

When θ is small, the arc looks like the opposite side

On a circle of radius R, a tiny angle makes opposite ≈ arc = Rθ, and adjacent ≈ R.

sin θ = P/H ≈ (Rθ)/R = θ
cos θ = B/H ≈ R/R = 1
tan θ = sin θ / cos θ ≈ θ / 1 = θ

θ must be in radian. Use only if θ ≤ 5°.

sin 2° is not 2. Convert 2° to radian first, then the approximation is that radian number.

Small-angle values

Using the small-angle approximation, find the value of each of the following.

(i) $\sin 2°$

(ii) $\tan 3°$

(iii) $\cos 3°$

Convert to radian, then approximate

$\sin 2° \approx 2°$ in radian $= 2 \times (\pi/180) = \pi/90$

$\sin 2° \approx \pi/90$

$\tan 3° \approx 3 \times (\pi/180) = \pi/60$

$\cos 3° \approx 1$

$\tan 3° \approx \pi/60$ $\cos 3° \approx 1$

Which value of tan 2° is correct?

Which of the following is correct for the value of $\tan 2°$ under the small-angle approximation?

1. $2°$
2. $\pi / 90$
3. $\pi / 30$
4. 2 rad

tan 2° is not 2°

Wrong: $\tan 2^\circ = 2^\circ$

Right: $\tan 2^\circ \approx 2 \times (\pi/180) = \pi/90$

$\tan 2^\circ \approx \pi/90$

The approximation replaces the ratio by the radian measure, never by the degree number.

Allied angles to finish

Simplify.

(13) $\sin(180° - \theta)$

(14) $\cos(180° + \theta)$

Copy the two lines. Expand with the compound formulas. Do not wait for the board.

Match the function to its description

Suggest a suitable match between the function in Column I and its description in Column II.

1. A → PT, B → QT, C → QT, D → PS
2. A → PT, B → QS, C → QT, D → PS
3. A → QT, B → QS, C → PT, D → PS
4. A → QS, B → PT, C → QT, D → PS

Column I	Column II
(A) sin 390°	(P) Positive
(B) sin (−30°)	(Q) Negative
(C) cos 120°	(R) Zero
(D) tan (−120°)	(S) Modulus is greater than one
	(T) Modulus is less than one

After class

Daily test before the next lecture.
Write the six compound-angle lines ten times on a rough page.

Tonight: the two allied simplifications and the match-the-column.

Leftover radian — now locked

180° = π rad

Wherever you see π rad, put 180°.

degree → radian: multiply by π/180
radian → degree: multiply by 180/π
Arc length = R θ with θ in radian. Never R × 90°.

90° on a circle of radius R → distance πR/2

What we lock from today's feel

If θ is not inside sin, cos or tan, convert it to radian first.

$$y = \sin\theta / \theta \quad \text{at} \quad \theta = 30° \quad \rightarrow \quad 3/\pi$$

$$\sin(A + B) = \sin A \cos B + \cos A \sin B$$

$$\cos(A + B) = \cos A \cos B - \sin A \sin B$$

$$\sin 2\theta = 2 \sin\theta \cos\theta \qquad \cos 2\theta = 1 - 2\sin^2\theta$$

Even, odd, and allied

$\cos(-\theta) = \cos\theta$ $\sin(-\theta) = -\sin\theta$ $\tan(-\theta) = -\tan\theta$

$\sin(90° - \theta) = \cos\theta$ $\sin(180° - \theta) = \sin\theta$

$\sin 15° = (\sqrt{6} - \sqrt{2})/4$ $\sin 75° = \sin 105° = (\sqrt{6} + \sqrt{2})/4$

Small angle — the one lock-line

$$\sin\theta \approx \theta \qquad \tan\theta \approx \theta \qquad \cos\theta \approx 1 \qquad (\theta \text{ in radian},\ \theta \le 5^\circ)$$

$\sin 2^\circ \approx \pi/90$ $\tan 3^\circ \approx \pi/60$ $\cos 3^\circ \approx 1$

Never leave the degree number in the approximation. Convert, then drop the word radian.

Daily test before the next lecture. Compound lines ten times tonight.